Activity 2 - Build a Significant GHG Source and Sink Register

Alignment: K1 | K5 | A3 | A7 | LO1 Duration: 60 minutes Objective: Classify direct and indirect GHG sources, sinks and reservoirs and assess their significance for reporting.

Scenario

EcoForge uses natural gas ovens, company vans, refrigerants, purchased electricity and rooftop solar. It buys aluminium and electronics, imports components, outsources warehousing, sells energy-consuming equipment, sends scrap for recycling and has staff travel and commuting. A landscaping contractor claims the site trees remove carbon, but no quantified evidence exists.

Materials

- Operational process narrative
- Energy, fleet and procurement summaries
- ISO 14064-1 category prompts
- Significance scoring worksheet

Questions

- Which sources are direct and which are indirect by category?
- Which GHGs and removal claims require investigation?
- What significance criteria are defensible for EcoForge?
- Who owns each source record and what primary evidence exists?

Detailed procedure

1. Walk through the value chain from purchased materials to product end of life and list every plausible source, sink and reservoir.
2. Classify sources into direct emissions, imported energy, transportation, products used, use of products and other indirect emissions.
3. Identify relevant gases such as carbon dioxide, methane, nitrous oxide and refrigerants without assuming carbon dioxide alone.
4. Define significance criteria using magnitude, influence, risk, stakeholder interest, data quality and reduction opportunity.
5. Score each indirect source and record the evidence or judgement supporting the score.
6. Select significant indirect sources for inclusion and document transparent reasons for any exclusion.
7. Assign a data owner, data source, frequency and evidence type to each included source.
8. Assess the landscaping removal claim and state what quantification and permanence evidence would be needed.
9. Run a completeness challenge using utility bills, ledger accounts, facility walk-through and procurement categories.

Required evidence

- Completed source/sink register
- Significance criteria and scores
- Inclusion/exclusion log
- Data-owner matrix

Acceptance check

The register contains all scenario sources, uses the six-category logic consistently, treats removals separately from offsets and gives evidence-based significance decisions.

Files

- Activity-02-Workbook.xlsx - editable working file
- Activity-02-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

Use fictional, anonymised or authorised workplace information only.